Arimoclomol for BUB1B-MVA: a functional rescue proposal

Track 2 · adriamasnou · 24 October 2026
Research proposal. No drug rescue experiment or treatment in this child has been performed.

The proposed candidate and the decision it must earn

We propose **arimoclomol as a conditional preclinical repurposing candidate** for an abundance-limited, functionally rescuable *BUB1B* phenotype. The hypothesis is that augmenting a cellular stress response could increase the pool of functional BubR1. Three links remain unproven: the effect of **p.Asn1002Lys (N1002K)**, the drug's effect on this protein, and rescue at relevant exposure. Our proposed experiment tests those links separately. **[OUR INFERENCE]**

The candidate has a defined human regulatory and pharmacokinetic record, but no established MVA indication. Its US indication is neurological manifestations of Niemann–Pick type C, **with miglustat**, from age two; its clinical mechanism in NPC remains unknown. This is a reason to examine translation carefully, not evidence of efficacy or safety in this child. MIPLYFFA label, revised March 2026, §§1, 12.

Delivered now: a source-traceable **gene-to-condition navigator**, an **interpretation canary**, an executable **rescue-decision prototype** and reproducible public-data evaluations. **Proposed next:** measure full-length N1002K function and compare drug effects. The reviewer walkthrough provides local examples and expected outputs; the laboratory programme remains unperformed.

1. Genotype and mechanism: distinguish evidence from inference

Our analysis identified heterozygous *BUB1B* **c.2210T>G, p.Leu737*** and **c.3006T>G, p.Asn1002Lys**. Their location on opposite parental alleles is **inferred, not established** by parental genotyping or read-based phase. The nonsense allele is predicted to undergo nonsense-mediated decay by analogy to nearby truncations; decay and residual protein have not been measured for this allele. We therefore do not assert that every BubR1 molecule in this child is N1002K. Track 1 analysis; PMID 20516114. **[OUR MEASUREMENT; OUR INFERENCE]**

Published work separates protein amount from function. I909T and L1012P reduce abundance; forced expression restored their nocodazole-induced checkpoint response in the tested system, not every mitotic function. Q921H retained approximately normal abundance and WT-like performance in the reported cellular alignment, segregation and checkpoint assays. Neither its fragment thermal behaviour nor clinical benignity follows from those results. A domain location alone cannot establish instability. Neither observation classifies N1002K. PMID 20516114.

Human BubR1's C-terminal domain has structural functions beyond any catalytic interpretation. Stability and chromosome alignment can be disrupted through different mechanisms; the domain contributes to PP2A-B56 recruitment and checkpoint silencing. We do not base treatment selection on a prediction of lost kinase activity. PMID 22698286; PMID 33207204.

Working categories are abundance-limited, function-limited, mixed and indeterminate. They are experimental outcomes, not diagnoses assigned from a structure prediction. A normal melting temperature of an isolated fragment cannot exclude a full-length stability, degradation, localisation or interaction defect. A rise in total protein does not by itself establish rescue. **[OUR INFERENCE]**

Local annotation retained all 313,417 distinct input variants. A prespecified phenotype-blind scan placed *BUB1B* in a severity-tied block at ranks 53–61 among 230 genes. This has limited discrimination. Low local SpliceAI scores, with positive controls, do not experimentally exclude splice effects. Annotation methods; splice assessment. **[OUR MEASUREMENT]**

2. Why arimoclomol merits a bounded test—and what could disqualify it

The primary pharmacology is context-dependent. Kuta and colleagues observed arimoclomol-associated Hsp70 induction in stressed murine neuronal cultures using low-micromolar exposures; mutant-FUS neurons did not show the same induction. These are neuronal stress models, not dividing N1002K cells. The result supports testing a conditional mechanism and argues against assuming a uniform response. PMID 31900865, Figures 5–7 and Online Resource 2. The supplement tests 1–10 μM and a 4 μM time course: the Hsp70 response at 6–24 hours was absent at 48 hours. An isolated late measurement could miss an earlier response.

A separate primary study reported improved 24-hour survival with arimoclomol in chemically stressed rat ventral-horn cultures, including a 1 μM condition. Celastrol produced Hsp70 staining alongside injury in that study. This supports pairing pathway measurements with function and cell fate; neither neuronal survival nor Hsp70 induction demonstrates mitotic rescue. PMID 19183864, Figures 1–3. **[OUR INFERENCE]**

NPC findings conflict even within the same GM18453 I1061T/I1061T line. Pipalia found no cholesterol-storage correction in the displayed 50–500 μM experiment (PMID 34481829); Shammas later reported reduction with prolonged treatment at 100–200 μM (PMID 40215728). Duration, formulation and culture conditions differ, but those differences have not been shown to explain the conflict. Much of the later mechanism work uses 400 μM; it does not establish BubR1 rescue or reachable exposure. Relevant company interests are disclosed. Motor-neuron work also separates selected functional effects from uniform HSP induction (PMID 38570009). **[OUR INFERENCE: transfer limits]**

Regulatory evidence is mixed. EMA recommended refusing Meplyffa's NPC application on 23 July 2026 because efficacy was insufficiently demonstrated. Its 31 July update records a re-examination request; the October draft agenda lists that procedure, not a final outcome. EMA product page; October agenda, p.12. Separately, its earlier assessment describes HSP70/NPC1 effects at concentrations including 400 μM and questions concentration-response. Earlier assessment, pp.21–22. These NPC findings constrain extrapolation; they do not test MVA efficacy. **[OUR INFERENCE: transfer limits]**

Clinical translation is a material counterargument: the larger ALS and inclusion-body myositis (IBM) trials did not demonstrate efficacy. Active arms had more adverse-event discontinuations and hepatic signals; one IBM participant had renal inflammation/injury considered treatment-related. Higher exposure cannot simply be assumed to solve a negative experiment. These adult disease-specific results do not test BUB1B or establish an MVA contraindication. PMID 38782015; PMID 37739573. The positive 2021 NPC trial is disease-specific (PMID 34418116). Its 2025 rescored-endpoint analysis uses the same trial, not independent replication (PMID 40520915). **[OUR INFERENCE: transfer limits]**

The label estimates paediatric NPC steady-state mean serum trough/peak concentrations of **206/523 ng/mL**, equivalent to **0.66/1.67 μM** using free-base molecular weight 313.78 g/mol. Serum values do not establish intracellular free concentration or sustained peak exposure. Label §§11–12; reproducible calculation. **[OUR MEASUREMENT]**

Require functional benefit at qualified exposure. The proposed initial series is vehicle, 0.2, 0.6, 1 and 2 μM; 4 and 10 μM are exploratory. These are laboratory design values, not human doses. Measure exposure and account for binding and time course. A high-exposure-only effect fails the translational gate. **[PROPOSED EXPERIMENT]**

3. The experiment: four gates with explicit failure outcomes

Gate A — establish what the variant does. Compare sequence-verified full-length WT and N1002K in a controlled replacement system, with endogenous BubR1 addressed consistently. Include empty/depleted and WT-rescue controls; I909T or L1012P supplies a published low-abundance/checkpoint-rescue comparator. It does not calibrate every functional endpoint. Measure transcript, protein abundance, turnover and localisation. Titrate WT to match N1002K abundance before comparing function: otherwise a functional deficit could simply reflect unequal amounts. Confirm any result in an independent preparation and ultimately a patient-relevant system. **[PROPOSED EXPERIMENT; design informed by PMID 20516114]**

Gate B — validate measurement and engagement. Qualify linear protein measurement, dynamic range and repeatability before testing efficacy. Use HSPA1A and other justified pathway readouts alongside BubR1. Validate normalisers under each condition; total-protein normalisation is a candidate, not an automatic guarantee. Include a separate positive control for the stress-response assay. Biological variation in a clinical PBMC study is not analytical CV for a cell assay: the former cannot size the latter. PMID 34418116. **[PROPOSED EXPERIMENT]**

Gate C — require function, not a protein-only hit. Quantify chromosome alignment and segregation errors per completed division, checkpoint response, division timing, viability and proliferation. The error rate among completed divisions is conditional: also report starting cells, mitotic entry, completion, death, arrest and censoring, each with its denominator. Advancement requires prespecified division-rate and fate bounds so selective loss cannot be ignored. Use blinded scoring and randomized plate positions. Plan biological replicates from pilot variance and a prespecified worthwhile effect; lock functional timing and dose–time multiplicity, and report confidence intervals and every tested concentration. **[PROPOSED EXPERIMENT]**

Gate D — interpret each outcome correctly. Define benefit Δ as the reduction in segregation-error probability and register the smallest worthwhile benefit δ before confirmation. A multiplicity-adjusted confidence interval wholly above δ supports advancement; wholly below δ rules out that programme benefit; touching or crossing δ is inconclusive. Separate exposure, division-rate and fate criteria still apply. **[PROPOSED EXPERIMENT]**

- **Abundance rises and function improves at relevant exposure:** advance to independent replication and a patient-relevant model; this still does not justify prescribing.
- **Abundance rises alone:** protein is an inadequate efficacy surrogate. Reject for insufficient efficacy only when the upper confidence bound $U < \delta$; an interval touching or crossing δ remains inconclusive.
- **Qualified assays and exposure give an upper confidence bound $U < \delta$:** reject the tested drug–exposure–system combination, whether or not HSP markers rise. A nonsignificant result alone does not meet this rule or disprove the class.
- **Function improves without the expected HSP response:** replicate the functional result, exclude selection, and investigate mechanism separately. This does not validate HSP-mediated BubR1 rescue. Even concordant HSP/protein/function changes require a controlled dependency experiment before claiming causal mediation.
- **Controls or exposure measurements fail:** label the experiment indeterminate and repair it; do not count it as a biological negative.
- **Only a fragment assay is negative:** retain the full-length functional question. Fragment thermal measurements are supporting evidence, not the decisive gate. **[OUR INFERENCE; PROPOSED EXPERIMENT]**

The laboratory package supplies verified insert designs, a staged functional pilot, sample/plate quantities, dated supplier prices and detailed quotation requests. Imaging feasibility, full costs, laboratory acceptance and timelines remain unconfirmed. Catalogue-priced materials are not an executable project budget. Functional protocol. **[ESTIMATE; PROPOSED EXPERIMENT]**

4. Alternatives and mechanistic cautions

HSP90 inhibition: tested destabilising BubR1 mutants lost protein under HSP90 inhibition. This supports a mechanistic caution when considering that axis, conditional on an untested allele's behaviour; it is not an established clinical contraindication for N1002K. PMID 20516114.

Fisetin: Salmela and colleagues observed substantial loss of kinetochore BubR1 and premature mitotic exit in dividing cells at 30 μM. We retain this as a **mechanistic caution against assuming an OTC senolytic is beneficial**. Human exposure at supplement doses and clinical harm in MVA were not established by that experiment. It does not establish the same effect for quercetin or dasatinib. PMID 19395653. A clinician or pharmacist should assess supplement use; the report does not advise a family to change treatment independently.

NAD precursors and rapamycin: North et al. directly measured increased BubR1 abundance after NMN in cultured MEFs and aged WT mouse testes; this is stronger target-protein evidence than we hold for arimoclomol. Lifespan extension was tested with a SIRT2 transgene, not NMN or NR. Mutant function, clinical exposure and safety remain unresolved. An NR implanted-tumour/metastasis-model signal warrants scrutiny without establishing NMN risk or this child's risk. Preliminary rapamycin observations are not a controlled published negative trial. PMID 24825348; PMID 36371959; PMID 23602569. **[OUR INFERENCE]**

4-PBA and TUDCA: the earlier "ER-only" exclusion is withdrawn. In CF epithelial cells, 1 mM 4-PBA transiently increased inducible Hsp70 through an Elp2/STAT3-associated pathway; it did not test BubR1. Hsp70 returned towards baseline by 24 hours while Hsc70 behaved differently. This provides a concrete reason to distinguish compartments, chaperones and sampling times. It establishes neither clinical reachability nor rescue by 4-PBA or TUDCA. PMID 22069317, Figures 1–8. **[OUR INFERENCE]**

Functional comparators: UPCP reduced aneuploid metaphases in progeria fibroblasts, with attenuation after BUBR1 knockdown (PMID 37118121). Engineered enhancement of Plk1 or B56 binding rescued mitotic delay in a different BubR1 construct (PMID 39516273). We independently recomputed selected published source-data summaries, retaining discrepancies elsewhere. These support assay controls and comparator hypotheses; neither study measures N1002K rescue, arimoclomol benefit or comparative MVA efficacy. UPCP is an experimental peptide, not an established repurposed MVA therapy. **[OUR REANALYSIS; OUR INFERENCE]**

Selection rule: arimoclomol is a conditional lead, not a demonstrated winner. If the model shows rescuable low abundance, compare vehicle, arimoclomol, exposure-qualified 4-PBA and research-only NMN using common functional endpoints and genetic controls. Estimate treatment differences with confidence intervals; separate significance against vehicle does not establish superiority. No candidate advances without passing exposure and fate criteria. A replicated cellular result then needs a justified tissue-output test aligned with the intended benefit. Candidate comparison and switch criteria; functional protocol. **[PROPOSED EXPERIMENT]**

5. Innovation that another team can use

Gene-to-condition navigator. The offline lookup maps the August 5 HFEA register through ClinVar, HPO and gene-MIM records: **2,063 rows, 2,028 mapped, 35 unmapped, 1,782 gene symbols**. It retains source/OMIM evidence and nine ClinVar/HPO disagreements. A match does not establish eligibility; no match does not establish ineligibility. Provenance; builder. **[OUR MEASUREMENT]**

The tool sends no queries and links to the regulator. Its 28-query audit reviewed all 47 returned gene–row associations: all 20 positive queries recovered expected rows, versus five with literal title matching; 11 omitted source qualifiers were repaired. Public controls demonstrate operation beyond MVA. This unblinded evaluation shares the tool's source data; clinical accuracy, superiority to the live HFEA service and family/clinician outcomes remain unmeasured.

Interpretation canary. The portable panel rejects missing, malformed or nonfinite outputs. Real SpliceAI inference passes six controls; its separate checker rejects nine injected failures. These are regression checks, not clinical validation. Documented outputs exclude patient-scale data and reference genomes. **[OUR MEASUREMENT]**

The rescue-decision prototype executes functional, exposure, fate and uncertainty rules: 40 synthetic cases matched expected decisions; seven faulty rule versions were detected. A separate 720,000-experiment simulation reproduced exactly under declared assumptions. Neither evaluates drug rescue or clinical performance. **[OUR MEASUREMENT]**

Executed public evaluation. Across 96 genomic jobs, original-HPO top-1/5/10 recovery was **6/7/8 (HG002) and 7/8/8 (HG001) of nine source labels**; five carry material qualifications. Phenotype-only recovered **4/8/8** in a different candidate universe. COG8/SLC35C1 no-HPO best modes use only one of two source alleles. Evaluation and outputs; FGFR2 reconstructions. These semi-synthetic checks demonstrate execution and ranking, not clinical accuracy. **[OUR MEASUREMENT]**

A ten-hypothesis benchmark completed 1,418 jobs across 713 source signatures. Autosomal top-1 was **556/681 with original HPO versus 430/681 without**, after a disclosed parser correction. The post-inference audit records 282 full-main-text reads, including 162 complete paper reviews. A separately reproduced source-quality analysis retains all 162 categories, including adverse results. Both audits are non-blinded; source-label recovery is not clinical accuracy. **[OUR MEASUREMENT]**

6. Next steps and Claude Science credits

Claude Science credits. The conditional $12,000 plan allocates $3,600 to evidence checks, $3,000 to public validation, $2,400 to analysis tools, $600 to service qualification, $1,200 to reuse and $1,200 to reserve. Cost/acceptance checks govern spending; laboratory funding is separate. **[PROPOSED WORK; award terms pending]**

The decisive test remains **full-length N1002K function and arimoclomol rescue at relevant exposure**, followed by independent replication and clinical review if positive.

Acknowledgement. This work was made possible through the Hackathon, organized by Sage Bionetworks in partnership with the MVA Society, Hugging Face, and BEACON (The Benchmarking, Evaluation, and Assessment Consortium for Science), with prize sponsorship from AWS and Anthropic. We are deeply grateful to the child and their family who generously contributed their data and their story to advance research into this rare disease. We acknowledge their trust in making this Hackathon possible.